

# EFPB Research Pipeline

Experiment run report: calibration, freeze and readiness

Runs of 19 September 2026 on one laptop (AMD Ryzen 7 5800U, 8 cores / 16 threads, 30 GB RAM, Ubuntu Linux)

## 1. Summary

- **The experiment runs on one computer** (8 parallel workers); the two-computer split in the instructions only divides the work. Both backends work end to end: the surrogate and the real SUMO/TraCI runner with virtual pedestrians.
- **Eleven defects were found** on the way, ten fixed and one worked around (section 3), among them a wrong SUMO signal program that made every SUMO controller drive the wrong phases, missing yellow/all-red clearance, a parallel-run port collision, a surrogate that served more than its configured capacity, and waits that ignored people still queued at the end (which made starving a mode look good). One of them (the split-scaling bug, defect 9) was introduced by me during calibration and caught before it was used.
- **Calibration is done and frozen for both backends** (sections 4 and 5): two demand levels near and above saturation were added, every controller was tuned under the same budget (30 configurations), the picks were confirmed on fresh seeds, and the frozen parameters and a preregistration draft are in the repository. In the surrogate only efpb changes from its defaults; in SUMO every controller keeps its defaults.
- **The traffic-simulation experiments have been run** (section 6): 1,920 surrogate runs and 520 SUMO runs with the frozen parameters. In the nominal cells efpb has the lowest person delay in SUMO and has a lower maximum wait than efficiency (about equal to equal\_bargaining), but the tuned fixed-time plan has the lowest maximum wait; under saturation there is no clear ordering and equal\_bargaining collapses. They were started without confirmation that the preregistration draft had been registered first, so treat them as exploratory unless it was. All results are gathered in the companion document **EFPB\_Simulation\_Results.pdf**; the 30-seed SUMO run is kept as future work.
- **What does not exist yet:** the HCI participant study and the preregistered statistical models, the ablation variants, SUMO robustness scenarios and real pedestrians (section 9). Two scenarios are unusable as implemented and are noted in section 6.

## 2. Running on one computer

Each process runs one job at a time, so the speed-up comes from starting several processes, each taking a different shard. Measured on this laptop with 8 workers:

| Run                                                                       | Runs  | Wall time    |
|---------------------------------------------------------------------------|-------|--------------|
| Surrogate calibration selection (80 configurations, lean mode)            | 3,200 | about 23 s   |
| Surrogate confirmation and delta sensitivity (fresh seeds)                | 2,400 | about 22 s   |
| SUMO calibration selection (80 configurations, all controllers)           | 2,800 | 34 min 36 s  |
| SUMO confirmation and delta sensitivity (fresh seeds)                     | 700   | 8 min 42 s   |
| SUMO pilot (4 controllers x 3 scenarios x 3 seeds)                        | 36    | about 50 s   |
| Surrogate matrix: confirmatory_core, robustness, stress (trace-free mode) | 1,920 | 65 s         |
| SUMO experiment: 13 cells x 4 controllers x 10 seeds, 4,200 s each        | 520   | 17 min 21 s  |
| Old 1,540-run surrogate matrix (earlier simulator versions, now stale)    | 1,540 | 58 to 73 min |

The instructions' estimate of 6.4–8.6 hours on two computers assumed 20 s per run; per-run cost differs by about 30 times between controllers. A full-length SUMO matrix has not been timed. Lean mode (no traces or per-second files) makes surrogate runs about 500 times faster and gives identical metrics.

### 3. Defects found and fixed

| #  | Defect                                                                                                                                                                | Effect                                                                                                                                            | Fix                                                                                                                                  |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| 1  | SUMO signal program: <code>cross.tls.add.xml</code> defines a second program that SUMO activates; the runner's phase numbers belonged to the network's own program 0. | In the first pilot V_NS was almost all red and V_EW fully red; every controller drove the wrong phases.<br>equal_bargaining completed 0 vehicles. | Program 0 is selected; approach pairs 1+2 and 3+4 match the signal.                                                                  |
| 2  | SUMO ignored vehicle demand; the controller was told the vehicle arrival rate was 0.                                                                                  | The 'vehicle heavy' scenario equalled the balanced one.                                                                                           | Bundled flows are rescaled to the configured rate; the rate is passed to the controller.                                             |
| 3  | SUMO metrics: wait counted time since first seen, warm-up was included, phase switches counted re-issued phases, pedestrians served counted phase steps.              | P95 waits above the run length; identical switch counts for every controller.                                                                     | Waits use consecutive stopped time; metrics cover the evaluation window; real counts.                                                |
| 4  | No yellow / all-red clearance between phases (SUMO and surrogate).                                                                                                    | 79 emergency stops in the 36 SUMO pilot runs. The surrogate charged no lost time for switching.                                                   | yellow 3 s and all-red 1 s in both backends: 0 emergency stops, 0 collisions.                                                        |
| 5  | Surrogate metrics covered the warm-up; the surrogate controller received arrival rates of 0.                                                                          | Throughput about 16% too high.                                                                                                                    | Metrics follow arrivals after warm-up; nominal rates are passed (surrogate 0.2.0).                                                   |
| 6  | Parallel SUMO runs could pick the same TraCI port; one client could read the other run's simulation.                                                                  | 1 of 36 regenerated runs differed; found by a bind error in its log.                                                                              | Serialised or per-worker ports plus a check of seed, scale and log path after connecting.                                            |
| 7  | Surrogate capacity was rounded up (0.8 to 1 vehicle per second per approach), and the demand generator capped arrivals at 0.95 per second.                            | The surrogate served 25% more than configured and could not produce saturating demand.                                                            | Fractional service credit; arrival cap lifted; existing demand streams unchanged (surrogate 0.3.0).                                  |
| 8  | Waits ignored people still queued at the end of a run.                                                                                                                | A controller that starved a mode looked good on max wait (equal_bargaining showed a pedestrian wait of 0 with 0% served).                         | Unserved people count with the wait so far (surrogate 0.4.0, SUMO runner 0.5.0).                                                     |
| 9  | Static fixed-time splits were scaled by the config's cycle (37) instead of their own total (90).                                                                      | In the new sweep one vehicle phase was never served, so a broken plan won the first selection.                                                    | Scale by the splits' own total and refuse a phase with no time. Caught before use; the first selection was discarded.                |
| 10 | The default fixed-time baseline (static, 90 s) was untuned.                                                                                                           | The research plan forbids comparing a tuned proposal with an untuned default.                                                                     | Demand-based plan added and calibrated; a 37 s cycle wins in both backends (section 5).                                              |
| 11 | Surrogate scenario S11 (accessibility): the accessibility flag stays on for the whole run, which marks every vehicle plan unsafe.                                     | Vehicles are never served by any controller (0% served, about 1,440 s person delay).                                                              | Not fixed (pedestrian walking time is not modelled). Excluded from the analysis; the reason follows from the code, not the outcomes. |

Still open: scenario S7 (emergency) is a single-second request with no visible effect and S12 (sensor error) makes every controller use the same fallback plan, so both are valid but uninformative; the instructions' two-computer commands skip the ablations, their runtime estimate is out of date, the pedestrian clearance interval is not modelled, and the SUMO runner ignores the emergency, sensor-error, accessibility and bursty scenario fields.

### 4. Demand levels closer to saturation (step 2)

A ladder of demand levels (multiples of the previous top level X: 0.15 pedestrians and 0.44 vehicles per second) was run with every controller at its then-current defaults, on calibration seeds only. Cells show the worst service ratio (share of arrivals served) and the largest wait divided by its 100 s limit (people still queued count with the wait so far).

**Surrogate** (worst service ratio / max wait per limit):

| x X | ped / veh per s | fixed_time  | efficiency  | equal_bargaining | efpb        |
|-----|-----------------|-------------|-------------|------------------|-------------|
| 0.5 | 0.075 / 0.22    | 0.99 / 0.29 | 0.98 / 0.54 | 0.99 / 0.56      | 0.97 / 0.48 |

| x X | ped / veh per s | fixed_time  | efficiency  | equal_bargaining | efpb        |
|-----|-----------------|-------------|-------------|------------------|-------------|
| 1.0 | 0.150 / 0.44    | 0.98 / 0.38 | 0.98 / 0.53 | 0.98 / 0.53      | 0.98 / 0.53 |
| 1.5 | 0.225 / 0.66    | 0.98 / 0.53 | 0.97 / 0.57 | 0.96 / 1.30      | 0.97 / 0.55 |
| 2.0 | 0.300 / 0.88    | 0.87 / 1.53 | 0.92 / 1.80 | 0.83 / 3.54      | 0.66 / 3.33 |
| 2.5 | 0.375 / 1.10    | 0.66 / 2.96 | 0.63 / 6.43 | 0.00 / 8.99      | 0.62 / 6.49 |
| 3.0 | 0.450 / 1.32    | 0.56 / 4.07 | 0.47 / 8.99 | 0.00 / 8.98      | 0.56 / 7.51 |
| 3.5 | 0.525 / 1.54    | 0.46 / 5.08 | 0.35 / 8.99 | 0.00 / 9.00      | 0.53 / 7.49 |
| 4.0 | 0.600 / 1.76    | 0.39 / 5.65 | 0.25 / 8.99 | 0.00 / 8.99      | 0.32 / 8.69 |

**SUMO** (share of vehicle demand completed per hour / mean vehicles queued):

| x X | veh per s | fixed_time  | efficiency  | efpb        |
|-----|-----------|-------------|-------------|-------------|
| 1.0 | 0.44      | 1.01 / 3.8  | 1.00 / 3.3  | 1.00 / 3.4  |
| 1.5 | 0.66      | 0.88 / 19.7 | 0.91 / 18.5 | 0.90 / 19.2 |
| 2.0 | 0.88      | 0.69 / 29.4 | 0.70 / 30.8 | 0.67 / 36.7 |
| 2.5 | 1.10      | 0.58 / 39.2 | 0.52 / 44.3 | 0.49 / 46.2 |
| 3.0 | 1.32      | 0.48 / 39.8 | 0.26 / 57.7 | 0.25 / 57.6 |

- The old top level X is about half of capacity, which is why the controllers barely differed before. The surrogate saturates between 1.5x and 2x X; SUMO earlier, its vehicle throughput plateauing near 2,200 vehicles per hour (about 0.62 per second).
- **New levels** (in configs/design\_base.yaml, existing levels unchanged): **S** = 1.5x X (0.225 pedestrians, 0.66 vehicles per second), near saturation, and **O** = 2.0x X (0.30 and 0.88), over saturation. Existing configs and the fingerprints of stored runs are untouched.
- A stress experiment (configs/experiments/stress.yaml: cells S/S, O/O, H/S, S/H, 30 seeds) was added for the confirmatory phase.

## 5. Calibration and freeze (steps 1 and 3)

### 5.1 Policy, decided before any confirmation run

- **Equal tuning budget.** Every controller gets at most 30 configurations from its own search space, defaults always included; spaces of 30 or fewer are searched exhaustively, larger ones by seeded random search. fixed\_time: plan family and cycle (14). efficiency: horizon, service duration, switch cost (27). equal\_bargaining: horizon, service duration (9; its equal weights are by definition not tuned). efpb: horizon, service duration, delta, kappa, eta (30 of 180). Safety timings (minimum green, yellow, all-red) are not tuned.
- **Objectives** over everyone who arrived after warm-up: person delay (occupancy-weighted mean wait) and max normalised wait (largest wait / 100 s). Scenarios weighted equally: surrogate MM, LH, HL, HH, XX, SS, OO and bursty HH; SUMO the same without the bursty cell. Selection seeds 301–305.
- **Selection rule:** per controller, scale both objectives to 0–1 within that controller and take the configuration closest to the ideal point; an exact tie keeps the default; in SUMO a configuration with a collision is excluded. The picks are written to selection.json **before** any confirmation run and cannot be re-picked.
- **Adoption rule** on fresh seeds (surrogate 321–350, SUMO 361–370): a pick replaces the default unless it is worse in both objectives (paired, identical scenarios and seeds). Delta is also run over the plan's set {0, .02, .05, .10, .20}.

### 5.2 surrogate results

**Selection stage** (seeds 301–305):

| Controller       | Configs | Delay range (s) | Max wait range | Default (delay / max wait) | Pick                                                | Adopted? |
|------------------|---------|-----------------|----------------|----------------------------|-----------------------------------------------------|----------|
| fixed_time       | 14      | 18.6–38.2       | 0.50–1.46      | 18.6 / 0.50                | default                                             | yes      |
| efficiency       | 27      | 17.1–17.8       | 0.70–0.86      | 17.1 / 0.73                | default                                             | yes      |
| equal_bargaining | 9       | 21.2–23.7       | 0.99–1.55      | 21.2 / 0.99                | default                                             | yes      |
| efpb             | 30      | 17.6–23.2       | 0.70–1.27      | 18.2 / 0.87                | eta 2, kappa 4, delta 0.05, horizon 60, duration 15 | yes      |

**Confirmation on fresh seeds** (defaults, picks and efpb's delta set):

| Configuration confirmed on fresh seeds                            | Controller       | Delay (s) | Max wait | Worst scenario |
|-------------------------------------------------------------------|------------------|-----------|----------|----------------|
| horizon 60, switch cost 5, duration 25 (default)                  | efficiency       | 16.4      | 0.68     | 1.43           |
| eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default) | efpb             | 17.2      | 0.73     | 2.21           |
| eta 2, kappa 4, delta 0.05, horizon 60, duration 15               | efpb             | 17.4      | 0.69     | 1.92           |
| efpb delta 0                                                      | efpb             | 17.4      | 0.69     | 1.88           |
| efpb delta 0.02                                                   | efpb             | 17.4      | 0.69     | 1.88           |
| efpb delta 0.1                                                    | efpb             | 17.3      | 0.69     | 1.91           |
| efpb delta 0.2                                                    | efpb             | 17.3      | 0.69     | 1.93           |
| horizon 60, duration 25 (default)                                 | equal_bargaining | 19.5      | 0.99     | 3.59           |
| cycle 37, plan demand_based (default)                             | fixed_time       | 18.2      | 0.51     | 1.52           |

**Frozen** (configs/frozen\_surrogate.yaml): {"signal": {"fixed\_cycle\_s": 37, "fixed\_timing": "demand\_based"}, "per\_controller": {"efpb": {"controller": {"debt\_eta": 2, "debt\_kappa": 4}, "signal": {"service\_duration\_s": 15}}}}

| delta | Person delay (s) | Max wait | Max wait vs picked |
|-------|------------------|----------|--------------------|
| 0     | 17.38            | 0.6865   | -0.0050 ± 0.0045   |
| 0.02  | 17.38            | 0.6865   | -0.0050 ± 0.0045   |
| 0.05  | 17.37            | 0.6915   | +0.0000 ± 0.0000   |
| 0.1   | 17.33            | 0.6874   | -0.0041 ± 0.0022   |
| 0.2   | 17.34            | 0.6863   | -0.0053 ± 0.0028   |

### 5.3 SUMO results

**Selection stage** (seeds 301–305):

| Controller       | Configs | Delay range (s) | Max wait range | Default (delay / max wait) | Pick                                                | Adopted?          |
|------------------|---------|-----------------|----------------|----------------------------|-----------------------------------------------------|-------------------|
| fixed_time       | 14      | 20.4–46.7       | 0.50–1.75      | 20.4 / 0.50                | default                                             | yes               |
| efficiency       | 27      | 16.9–17.4       | 0.49–0.53      | 16.9 / 0.50                | horizon 30, switch cost 0, duration 25              | no (kept default) |
| equal_bargaining | 9       | 30.3–44.4       | 2.16–2.42      | 30.3 / 2.16                | default                                             | yes               |
| efpb             | 30      | 16.0–39.1       | 0.35–1.36      | 16.8 / 0.42                | eta 2, kappa 2, delta 0.05, horizon 90, duration 35 | no (kept default) |

**Confirmation on fresh seeds** (defaults, picks and efpb's delta set):

| Configuration confirmed on fresh seeds                            | Controller | Delay (s) | Max wait | Worst scenario |
|-------------------------------------------------------------------|------------|-----------|----------|----------------|
| horizon 30, switch cost 0, duration 25                            | efficiency | 16.7      | 0.51     | 0.86           |
| horizon 60, switch cost 5, duration 25 (default)                  | efficiency | 16.6      | 0.51     | 0.88           |
| eta 1.0, kappa 2.0, delta 0.05, horizon 60, duration 25 (default) | efpb       | 17.3      | 0.44     | 0.58           |

| Configuration confirmed on fresh seeds              | Controller       | Delay (s) | Max wait | Worst scenario |
|-----------------------------------------------------|------------------|-----------|----------|----------------|
| eta 2, kappa 2, delta 0.05, horizon 90, duration 35 | efpb             | 19.3      | 0.48     | 1.46           |
| efpb delta 0                                        | efpb             | 19.3      | 0.48     | 1.46           |
| efpb delta 0.02                                     | efpb             | 19.3      | 0.48     | 1.46           |
| efpb delta 0.1                                      | efpb             | 19.3      | 0.48     | 1.46           |
| efpb delta 0.2                                      | efpb             | 19.3      | 0.48     | 1.46           |
| horizon 60, duration 25 (default)                   | equal_bargaining | 30.6      | 2.23     | 8.99           |
| cycle 37, plan demand_based (default)               | fixed_time       | 24.1      | 0.62     | 2.52           |

**Frozen** (configs/frozen\_sumo.yaml): {"signal": {"fixed\_cycle\_s": 37, "fixed\_timing": "demand\_based"}}

| delta | Person delay (s) | Max wait | Max wait vs picked |
|-------|------------------|----------|--------------------|
| 0     | 19.29            | 0.4846   | +0.0000 ± 0.0000   |
| 0.02  | 19.29            | 0.4846   | +0.0000 ± 0.0000   |
| 0.05  | 19.29            | 0.4846   | +0.0000 ± 0.0000   |
| 0.1   | 19.29            | 0.4830   | -0.0016 ± 0.0015   |
| 0.2   | 19.33            | 0.4826   | -0.0020 ± 0.0023   |

## 5.4 The fixed-time cycle under stress

| Cycle (s) | Surrogate: delay / max wait / worst scenario | SUMO: delay / max wait / worst scenario |
|-----------|----------------------------------------------|-----------------------------------------|
| 37        | 18.6 / 0.50 / 1.53                           | 20.4 / 0.50 / 1.59                      |
| 40        | 20.9 / 0.60 / 1.92                           | 21.3 / 0.64 / 2.27                      |
| 45        | 21.7 / 0.62 / 1.95                           | 20.8 / 0.58 / 1.81                      |
| 50        | 19.8 / 0.54 / 1.30                           | 20.8 / 0.53 / 1.34                      |
| 60        | 22.5 / 0.62 / 1.44                           | 24.1 / 0.58 / 1.28                      |
| 75        | 24.6 / 0.67 / 1.26                           | 27.5 / 0.63 / 1.12                      |
| 90        | 27.1 / 0.74 / 1.21                           | 31.4 / 0.72 / 1.09                      |
| 120       | 34.3 / 0.95 / 1.36                           | 39.3 / 0.94 / 1.15                      |

Demand-based plans (splits from each scenario's nominal demand). Both backends pick 37 s, the shortest cycle that fits three 10 s minimum greens plus clearance. Longer cycles lower the worst-scenario wait (the over-saturated cell) but raise the overall delay, so the equal-weight rule keeps the short cycle. The static plans with proportional splits had a much higher max wait at every cycle length.

## 5.5 What the calibration shows, and what to be careful about

- **No controller is clearly ahead.** On the frozen configurations (fresh seeds): surrogate person delay / max wait – fixed\_time 18.2 / 0.51; efficiency 16.4 / 0.68; equal\_bargaining 19.5 / 0.99; efpb 17.4 / 0.69. SUMO – fixed\_time 24.1 / 0.62; efficiency 16.6 / 0.51; equal\_bargaining 30.6 / 2.23; efpb 17.3 / 0.44. The ranking differs by backend and by objective. This is calibration output, not a finding.
- **Delta barely matters.** Delta 0 and 0.02 give identical results in both backends (0.05 too in SUMO; in the surrogate it differs by under one percent), and 0.10 and 0.20 change the max wait by less than one percent. The default 0.05 is frozen; the full sensitivity set is reported above, as the plan requires.
- **equal\_bargaining collapses under saturation** and none of its tuning parameters helps: at the S and O levels pedestrians wait almost the whole run (max wait about 4 and 9 times the limit in SUMO). The cause is its individual-rationality rule, which never leaves a vehicle phase while vehicles are queued. Decide whether this is the intended baseline before the confirmatory runs; if it is not, comparisons with it will exaggerate any advantage of efpb.
- **The two backends freeze different parameters:** in the surrogate efpb changes (kappa 4, eta 2, service duration 15 s; a small gain in max wait for a small loss in delay), in SUMO every controller keeps its defaults because the picks did not survive the fresh seeds. The SUMO frozen set is the one to use for SUMO experiments; the surrogate is engineering validation only.

- **Equal weights make the over-saturated cell dominate** the averages. Per-scenario numbers are in the result files; a different weighting would be a deviation to preregister, not a tweak.

Max normalised wait by scenario for each controller's defaults (selection seeds; above 1 the limit is exceeded):

| surrogate        | K_MM | K_LH | K_HL | K_HH | K_XX | K_SS | K_OO | K_HH_burst |
|------------------|------|------|------|------|------|------|------|------------|
| fixed_time       | 0.29 | 0.30 | 0.29 | 0.29 | 0.38 | 0.53 | 1.53 | 0.42       |
| efficiency       | 0.54 | 0.86 | 0.55 | 0.49 | 0.53 | 0.57 | 1.80 | 0.50       |
| equal_bargaining | 0.47 | 0.47 | 0.49 | 0.50 | 0.53 | 1.30 | 3.54 | 0.62       |
| efpb             | 0.51 | 0.54 | 0.44 | 0.51 | 0.53 | 0.55 | 3.33 | 0.54       |

  

| SUMO             | K_MM | K_LH | K_HL | K_HH | K_XX | K_SS | K_OO |
|------------------|------|------|------|------|------|------|------|
| fixed_time       | 0.27 | 0.26 | 0.27 | 0.27 | 0.32 | 0.53 | 1.59 |
| efficiency       | 0.31 | 0.51 | 0.30 | 0.45 | 0.41 | 0.71 | 0.84 |
| equal_bargaining | 0.34 | 0.33 | 0.27 | 0.45 | 0.40 | 4.37 | 8.95 |
| efpb             | 0.30 | 0.46 | 0.27 | 0.43 | 0.39 | 0.56 | 0.56 |

## 5.6 Disclosures

- A first surrogate selection was made on flawed data (defect 9, and waits that ignored unserved people). I saw its confirmation results before discarding it; everything shown here was redone from scratch with the corrected code.
- An exact tie in the selection rule once changed a parameter for no measurable reason (efficiency's horizon 30 and 60 give identical results). The rule was refined so that ties keep the default; I saw that tie before changing the rule.
- Selection used the same seeds for all 80 configurations; the confirmation on fresh seeds is what guards against the optimism of picking the best of many.
- Nothing here is registered. docs/preregistration\_calibration.md is a draft generated from the frozen files; it only becomes a preregistration when you register it with a timestamp outside this repository, before the confirmatory runs.

## 6. Results of the matrix experiments

**What was run.** Surrogate: confirmatory\_core (9 demand cells x 4 controllers x 30 seeds), robustness (9 scenarios x 4 x 10; S11 excluded, so 8 analysed) and stress (4 cells x 4 x 30), 1,920 runs in trace-free mode (identical metrics). SUMO: the 9 nominal cells plus 4 stress cells x 4 controllers x 10 paired seeds, 4,200 s per run, 520 runs, virtual pedestrians. All with the frozen parameters. **These are descriptive results:** means over cells (cells weighted equally) and paired differences of efpb against each other controller on identical scenarios and seeds, with standard errors. There are no significance tests because the preregistered models do not exist yet. The runs were started without confirmation that the preregistration draft had been registered, so they are exploratory unless it was.

**SUMO/TraCI (virtual pedestrians): means over cells** (nominal 9, stress 4 cells).

| Group   | Controller       | Person delay (s) | Max wait / limit | Ped wait (s) | Veh wait or stop (s) | Collisions / e-stops |
|---------|------------------|------------------|------------------|--------------|----------------------|----------------------|
| nominal | fixed_time       | 8.5              | 0.27             | 11.2         | 7.7                  | 0.0 / 0.0            |
| nominal | efficiency       | 6.4              | 0.47             | 8.7          | 6.2                  | 0.0 / 0.0            |
| nominal | equal_bargaining | 6.7              | 0.42             | 6.0          | 6.8                  | 0.0 / 0.0            |
| nominal | efpb             | 6.3              | 0.41             | 8.7          | 6.0                  | 0.0 / 0.0            |
| stress  | fixed_time       | 48.6             | 1.48             | 72.3         | 37.6                 | 0.0 / 0.0            |

| Group  | Controller       | Person delay (s) | Max wait / limit | Ped wait (s) | Veh wait or stop (s) | Collisions / e-stops |
|--------|------------------|------------------|------------------|--------------|----------------------|----------------------|
| stress | efficiency       | 41.1             | 1.85             | 34.6         | 129.1                | 0.0 / 0.0            |
| stress | equal_bargaining | 338.9            | 18.04            | 807.5        | 324.7                | 0.0 / 0.0            |
| stress | efpb             | 42.0             | 1.54             | 23.5         | 133.8                | 0.0 / 0.0            |

**SUMO/TraCI (virtual pedestrians): efpb against each controller** (paired on scenario and seed; mean difference  $\pm$  standard error; in brackets the number of cells where efpb is lower; negative = efpb lower/better).

| Group   | efpb minus       | Person delay (s)          | Max wait / limit        |
|---------|------------------|---------------------------|-------------------------|
| nominal | fixed_time       | -2.18 $\pm$ 0.16 (9/9)    | +0.14 $\pm$ 0.01 (0/9)  |
| nominal | efficiency       | -0.13 $\pm$ 0.02 (9/9)    | -0.05 $\pm$ 0.01 (9/9)  |
| nominal | equal_bargaining | -0.36 $\pm$ 0.04 (9/9)    | -0.00 $\pm$ 0.01 (5/9)  |
| stress  | fixed_time       | -6.63 $\pm$ 9.99 (4/4)    | +0.06 $\pm$ 0.97 (1/4)  |
| stress  | efficiency       | +0.90 $\pm$ 13.91 (1/4)   | -0.31 $\pm$ 1.42 (4/4)  |
| stress  | equal_bargaining | -296.93 $\pm$ 79.45 (4/4) | -16.50 $\pm$ 2.37 (4/4) |

**Surrogate (engineering validation only): means over cells** (nominal 9, robustness 8, stress 4 cells).

| Group      | Controller       | Person delay (s) | Max wait / limit | Ped wait (s) | Veh wait or stop (s) | Starvation viol. |
|------------|------------------|------------------|------------------|--------------|----------------------|------------------|
| nominal    | fixed_time       | 12.3             | 0.30             | 10.9         | 12.7                 | 0.00             |
| nominal    | efficiency       | 9.9              | 0.70             | 12.3         | 10.0                 | 0.17             |
| nominal    | equal_bargaining | 10.1             | 0.57             | 8.9          | 10.5                 | 0.00             |
| nominal    | efpb             | 9.9              | 0.52             | 13.2         | 9.6                  | 0.00             |
| robustness | fixed_time       | 12.7             | 0.33             | 11.4         | 13.0                 | 0.00             |
| robustness | efficiency       | 11.0             | 0.59             | 12.4         | 10.8                 | 0.01             |
| robustness | equal_bargaining | 11.4             | 0.61             | 10.9         | 11.5                 | 0.27             |
| robustness | efpb             | 11.2             | 0.50             | 14.2         | 10.6                 | 0.00             |
| stress     | fixed_time       | 44.5             | 1.55             | 72.9         | 36.7                 | 575.06           |
| stress     | efficiency       | 36.1             | 1.55             | 29.4         | 38.2                 | 290.04           |
| stress     | equal_bargaining | 67.1             | 6.09             | 235.7        | 23.2                 | 361.59           |
| stress     | efpb             | 45.6             | 1.87             | 34.3         | 48.9                 | 396.49           |

**Surrogate (engineering validation only): efpb against each controller** (paired on scenario and seed; mean difference  $\pm$  standard error; in brackets the number of cells where efpb is lower; negative = efpb lower/better).

| Group      | efpb minus       | Person delay (s)        | Max wait / limit       |
|------------|------------------|-------------------------|------------------------|
| nominal    | fixed_time       | -2.42 $\pm$ 0.08 (9/9)  | +0.21 $\pm$ 0.00 (0/9) |
| nominal    | efficiency       | +0.00 $\pm$ 0.02 (5/9)  | -0.19 $\pm$ 0.01 (9/9) |
| nominal    | equal_bargaining | -0.19 $\pm$ 0.02 (8/9)  | -0.05 $\pm$ 0.01 (8/9) |
| robustness | fixed_time       | -1.47 $\pm$ 0.18 (5/8)  | +0.17 $\pm$ 0.01 (0/8) |
| robustness | efficiency       | +0.18 $\pm$ 0.05 (2/8)  | -0.09 $\pm$ 0.01 (7/8) |
| robustness | equal_bargaining | -0.20 $\pm$ 0.04 (6/8)  | -0.11 $\pm$ 0.02 (7/8) |
| stress     | fixed_time       | +1.13 $\pm$ 0.75 (2/4)  | +0.32 $\pm$ 0.08 (2/4) |
| stress     | efficiency       | +9.56 $\pm$ 1.47 (0/4)  | +0.32 $\pm$ 0.09 (3/4) |
| stress     | equal_bargaining | -21.45 $\pm$ 5.24 (3/4) | -4.22 $\pm$ 0.76 (4/4) |

**SUMO stress cells one by one** (S = near saturation, O = over saturation; cell T\_XY = pedestrian level X, vehicle level Y):

| SUMO stress cell | Metric           | fixed_time | efficiency | equal_bargaining | efpb |
|------------------|------------------|------------|------------|------------------|------|
| T_SS             | person delay (s) | 36.5       | 32.5       | 537.4            | 33.6 |
| T_SS             | max wait / limit | 0.54       | 0.85       | 26.76            | 0.58 |
| T_HS             | person delay (s) | 38.6       | 24.7       | 67.7             | 26.8 |
| T_HS             | max wait / limit | 0.41       | 1.26       | 8.06             | 0.57 |
| T_SH             | person delay (s) | 11.0       | 9.5        | 9.5              | 9.4  |
| T_SH             | max wait / limit | 0.54       | 0.46       | 0.49             | 0.45 |
| T_OO             | person delay (s) | 108.3      | 97.5       | 740.9            | 98.0 |
| T_OO             | max wait / limit | 4.45       | 4.85       | 36.85            | 4.58 |

## What the results show

- **Nominal demand, SUMO:** efpb has the lowest mean person delay (6.31 s against 6.45 for efficiency, 6.68 for equal\_bargaining and 8.49 for the tuned fixed-time plan) and is lower than each of them in all 9 cells (paired differences  $-0.13 \pm 0.02$ ,  $-0.36 \pm 0.04$  and  $-2.18 \pm 0.16$  s). Its max wait is lower than efficiency's ( $-0.05 \pm 0.01$ ) and about equal to equal\_bargaining's ( $-0.00 \pm 0.01$ ), but the tuned fixed-time plan has the lowest max wait (0.27 against 0.41 for efpb) at 2.2 s higher person delay.
- **Nominal demand, surrogate:** the same pattern. efpb ties efficiency on person delay (paired difference  $+0.00 \pm 0.02$  s) and its max wait is lower (paired difference  $-0.19 \pm 0.01$ ); the fixed-time plan again has the lowest max wait (0.30) at a higher person delay (12.3 s).
- **Under saturation there is no clear ordering.** In SUMO efpb, efficiency and the fixed-time plan are within about 7 s of each other on person delay and their standard errors are large; cell T\_OO (over saturation) fails for every controller (waits of several times the limit). In the surrogate efficiency has the lowest person delay under stress and efpb is higher than it by  $+9.56 \pm 1.47$  s (paired).
- **equal\_bargaining collapses under saturation** in both backends (SUMO stress: person delay 339 s, max wait 18 times the limit): pedestrians are almost never served while vehicles queue. This is a property of the baseline as implemented; whether it is the intended baseline is still open, and any advantage of efpb over it under stress should be read with that in mind.
- **Safety flags:** no collisions and no emergency stops in any SUMO run.
- **Robustness scenarios (surrogate):** S11 (accessibility) is excluded: the simulator keeps its flag on for the whole run, so vehicles are never served by any controller; it is a code defect, not an outcome. S12 (sensor error) makes every controller use the same fallback plan, and S7 (emergency) is a single-second request with no visible effect, so both are valid but uninformative. The remaining 6 scenarios behave like the nominal cells.

## Cautions

- The standard errors treat every scenario-and-seed pair as independent and ignore how different the cells are, so in the stress group (4 cells) they are not reliable; read the cell counts and the per-cell table instead.
- SUMO has 10 seeds per cell (the plan's design has 30) and virtual pedestrians; one intersection; the surrogate is engineering validation only.
- Max wait rewards a short fixed cycle by construction (every phase comes round within the cycle), so it favours the fixed-time plan; person delay favours adaptive control. Neither objective alone decides a winner.
- Nothing here says anything about explanations, comprehension or trust; that part of the research plan is untested.

## 7. The SUMO pilot (time-boxed, virtual pedestrians)

36 runs: 3 scenarios x 4 controllers x 3 seeds (9001–9003, never used for calibration). `fixed_time` uses the tuned demand-based 37 s plan. The pilot config keeps the provisional parameters (identical to the SUMO frozen set); all 36 runs completed with no collisions and no emergency stops. Means of 3 seeds.

| Scenario            | Controller       | Veh/h | Stop time (s) | Wait p95 (s) | Queue | Ped served | Ped wait (s) | Switches | Clear. % |
|---------------------|------------------|-------|---------------|--------------|-------|------------|--------------|----------|----------|
| S1_balanced         | fixed_time       | 792   | 7.5           | 18.7         | 1.63  | 65.0       | 10.4         | 73       | 18.7     |
| S1_balanced         | efficiency       | 783   | 7.3           | 16.7         | 1.55  | 65.3       | 9.0          | 72       | 18.1     |
| S1_balanced         | equal_bargaining | 788   | 8.3           | 20.3         | 1.76  | 63.3       | 6.1          | 72       | 17.0     |
| S1_balanced         | efpb             | 780   | 6.9           | 17.0         | 1.47  | 65.3       | 8.7          | 72       | 18.1     |
| S2_pedestrian_heavy | fixed_time       | 792   | 7.5           | 18.7         | 1.63  | 101.3      | 11.6         | 73       | 18.7     |
| S2_pedestrian_heavy | efficiency       | 789   | 8.5           | 17.3         | 1.81  | 99.7       | 8.5          | 73       | 17.6     |
| S2_pedestrian_heavy | equal_bargaining | 780   | 9.0           | 17.7         | 1.91  | 99.0       | 6.7          | 74       | 16.7     |
| S2_pedestrian_heavy | efpb             | 789   | 8.1           | 17.0         | 1.72  | 99.7       | 8.1          | 73       | 17.7     |
| S3_vehicle_heavy    | fixed_time       | 1239  | 9.4           | 19.0         | 3.02  | 65.0       | 10.4         | 73       | 18.7     |
| S3_vehicle_heavy    | efficiency       | 1219  | 7.4           | 17.7         | 2.34  | 64.7       | 13.2         | 72       | 20.0     |
| S3_vehicle_heavy    | equal_bargaining | 1236  | 9.0           | 22.7         | 2.83  | 64.7       | 6.5          | 73       | 17.9     |
| S3_vehicle_heavy    | efpb             | 1229  | 7.4           | 18.7         | 2.31  | 63.3       | 10.6         | 71       | 20.1     |

**Tuned fixed-time minus each adaptive controller** (same scenarios and seeds; mean  $\pm$  standard error, 9 pairs; negative = `fixed_time` lower):

| Metric                   | vs efficiency    | vs equal_bargaining | vs efpb          |
|--------------------------|------------------|---------------------|------------------|
| Vehicle stop time (s)    | +0.39 $\pm$ 0.47 | -0.63 $\pm$ 0.33    | +0.67 $\pm$ 0.39 |
| Vehicle wait p95 (s)     | +1.56 $\pm$ 0.34 | -1.44 $\pm$ 0.87    | +1.22 $\pm$ 0.55 |
| Pedestrian mean wait (s) | +0.54 $\pm$ 0.93 | +4.31 $\pm$ 0.26    | +1.65 $\pm$ 0.61 |
| Pedestrian max wait (s)  | -9.89 $\pm$ 3.25 | +2.78 $\pm$ 0.95    | -7.11 $\pm$ 2.46 |

- **Vehicles:** no controller is clearly ahead (stop times within about 1 s). **Pedestrians:** the trade-offs differ; `equal_bargaining` has the lowest pedestrian waits at these (light) demand levels. Only 3 seeds, and `fixed_time`'s vehicle results are identical in S1 and S2 (its timing ignores pedestrians), so the standard errors are optimistic. Treat this as descriptive.
- **Effect of clearance** (mean over the pilot; pilot with the old `fixed_time` plan, static 90 s, so that timing is not mixed in): mean vehicle stop time without clearance / with clearance – `fixed_time` 16.1 / 16.5 s; efficiency 4.6 / 7.7 s; `equal_bargaining` 5.5 / 8.8 s; `efpb` 4.4 / 7.5 s. Emergency stops: 79 without clearance, 0 with.
- Tuning the fixed-time plan (same seeds): mean vehicle stop time 16.5 s with the static 90 s plan against 8.1 s with the tuned plan.

## 8. Integrity and reproducibility

- **Versions and fingerprints.** Every stored run carries the simulator version (surrogate 0.4.0, SUMO runner 0.5.0) and a fingerprint of everything that decides its result (settings, scenario, controller, seed, plus the SUMO input files). A stored run made with another configuration or version is refused, not reused. Bump the versions by hand when a code change alters results.
- **Pre-committed selection.** `freeze_calibration.py` writes the picks before any confirmation run and refuses to re-pick if the selection results change.
- **Reproducibility.** Calibration, sweep and pilot outputs reproduced exactly when regenerated (checked before each version change); 49 automated tests pass, including a real-SUMO end-to-end test and

tests for most of the defects above.

- **Calibration-only data.** Calibration scenarios and seeds are refused if they overlap the confirmatory, robustness, ablation, stress or SUMO-pilot experiments.
- **Stale results are marked.** `data/RESULTS_STATUS.md` lists what is current and what is not. The old 1,540-run surrogate matrix predates surrogate 0.3.0/0.4.0, the frozen parameters and the stress levels; it must be moved aside before the matrix is run once.

## 9. Readiness for the confirmatory experiment

| Item                                                                   | Status             | What is missing                                                                                                                                                                                                         |
|------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pipeline, reproducibility, safeguards                                  | Ready              | –                                                                                                                                                                                                                       |
| Demand levels (S, O) and stress experiment                             | Done               | –                                                                                                                                                                                                                       |
| Tuning-budget policy, calibration, freeze (both backends)              | Done               | Register the preregistration draft; decide the weighting of scenarios if you want a different one.                                                                                                                      |
| Surrogate matrix (confirmatory_core 1,080, robustness 360, stress 480) | Done (exploratory) | Run trace-free in 65 s; S11 excluded. Engineering validation only. Rerun with traces only for the participant stimuli.                                                                                                  |
| Ablations (100 runs)                                                   | Not ready          | The ablated controllers are not implemented; the five cells are identical.                                                                                                                                              |
| equal_bargaining baseline                                              | Needs decision     | Collapses under saturation; confirm it is the intended baseline.                                                                                                                                                        |
| SUMO experiment (9 nominal + 4 stress cells)                           | Done (exploratory) | 10 seeds instead of 30 (the 30-seed run, about 50 more minutes, is kept as future work); virtual pedestrians. The robustness scenarios cannot be run in SUMO (runner ignores emergency, sensor, accessibility, bursty). |
| HCI participant study                                                  | Not started        | Dashboard, participant app, ethics approval, participants.                                                                                                                                                              |
| Statistical analysis and paper                                         | Not started        | Preregistered models (GLMM, LMM, ordinal) are not implemented; results table is still TBD.                                                                                                                              |

**Suggested next steps:** (1) register the preregistration (and say in the paper that the matrix runs preceded confirmation of it, if they did); (2) decide on equal\_bargaining; (3) implement the ablation variants and fix or drop scenario S11; (4) future work: rerun the SUMO experiment with 30 seeds (about 50 more minutes; the runner reuses the 10 seeds already stored) and decide the robustness scope; (5) implement the preregistered analysis; (6) start the ethics application and the participant study, which is the longest lead time.

## 10. Research-integrity note

Nothing in this report is a research finding. The surrogate is an approximation, the SUMO runs use virtual pedestrians, calibration output only chooses parameters, and no participant study has been carried out. Results from the first, invalid SUMO pilots and from the flawed first calibration were discarded, not used. All tables are generated from the stored result files by `docs/make_report.py`. The matrix runs were exploratory unless the preregistration was registered before they started.

## Appendix: where things are

| What                                  | Where (inside efpb-research-pipeline/)                                                                                                                                         |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Frozen parameters                     | configs/frozen_surrogate.yaml, configs/frozen_sumo.yaml                                                                                                                        |
| Preregistration draft                 | docs/preregistration_calibration.md (regenerate with make_preregistration.py)                                                                                                  |
| Demand levels, experiment configs     | configs/design_base.yaml; configs/experiments/{confirmatory_core, robustness, ablations, stress}.yaml                                                                          |
| Surrogate calibration                 | data/calibration_sweep/ (selection.json, summary.csv, freeze_report.md, confirmation/)                                                                                         |
| SUMO calibration                      | data/sumo_calibration/ (selection.json, summary_selection.csv, freeze_report.md, selection/, confirmation/)                                                                    |
| Saturation study                      | data/saturation_study/                                                                                                                                                         |
| SUMO pilot and comparison sets        | data/sumo_runs/, data/sumo_runs_static_fixed_time/, data/sumo_runs_v2_no_clearance/, data/sumo_fixed_time_check/                                                               |
| Stale or archived                     | data/runs/ (stale), data/runs_v1_no_clearance/, data/sumo_runs_v1_defective/, data/archive_*/                                                                                  |
| Matrix experiments and their analysis | data/matrix/ (surrogate runs), data/sumo_matrix/runs/ (SUMO runs), data/results/ (analysis CSVs). Scripts: run_sumo_matrix.py, logs/run_matrix_surrogate.sh, analyze_matrix.py |
| All results in one document           | ../EFPB_Simulation_Results.pdf (built by docs/make_results.py)                                                                                                                 |
| Status of every result                | data/RESULTS_STATUS.md                                                                                                                                                         |